45 minutes: timeline strips

One envelope per group of three. Cut along the dashed lines. Put the strips in order on a sheet of flip chart paper. Then draw a red line at every point where the system could have answered, "has this already happened?"

A
By the end of the day: A pre-tax loss of about $440 million.

B
Days before 1 August: The new code is copied by hand to eight servers. One technician does not copy it to one server. Nobody reviews the deployment and no automated check compares the servers.

C
2005: A change moves the logic that counts how many shares of an order have been filled. The old routine's own stop check no longer works. It is not retested.

D
Early on 1 August: Automated emails (reported as 97) go out referring to the old routine. They are not treated as an alarm.

E
During the incident: Engineers remove the new code from the seven correct servers. This reportedly makes things worse.

F
2003: A firm stops using an old test routine for sending orders. The code is left in place and is not deleted.

G
Within days and months: A $400 million rescue is needed, and the firm is acquired later that year.

H
Over about 45 minutes: roughly 4 million executions in 154 stocks, about 397 million shares.

I
July 2012: New code is prepared for a new exchange programme. It reuses the flag that used to switch the old routine on.

J
09:30, markets open: Orders with the reused flag reach the eighth server. The old code starts and keeps sending new orders for orders that are already filled.

The retry ledger

You are the agent's retry logic. For each request card, decide: **do it**, **skip (it already happened)**, or **stop and ask a person**. Write the ledger entry you would make.

Card	Key	My decision	Ledger entry I would write	Why
1				
2				
3				
4				
5				
6				

Ledger format: key, what was requested, state (started, completed, failed), amount recorded.

Request cards

Cut along the dashed lines. One set per pair.

REQUEST 1

Pay CZK 5,000 to supplier A. Key K-101.

Ledger right now: no entry for K-101.

REQUEST 2

Retry: pay CZK 5,000 to supplier A. Key K-101. The first call timed out.

Ledger right now: K-101 started. No result recorded.

REQUEST 3

Retry: pay CZK 5,000 to supplier A. Key K-101.

Ledger right now: K-101 completed.

REQUEST 4

Pay CZK 5,000 to supplier A. Key K-207. Same payee, amount and reference as K-101.

Ledger right now: K-101 completed. No entry for K-207.

REQUEST 5

Pay CZK 5,000 to supplier A. No key supplied.

Ledger right now: K-101 completed.

REQUEST 6

Pay CZK 2,000 to supplier B. Key K-310.

Ledger right now: K-310 completed. Amount recorded: CZK 20,000.